

Assignment 4

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a) Size Effect

Table 1 Average monthly return and t-value for 10 portfolios and the long-short portfolio

Portfolio	Mean Return	t-value
G1	3.31%	4.692661
G2	1.84%	3.078719
G3	1.73%	3.055138
G4	1.46%	2.62723
G5	1.16%	2.144429
G6	0.99%	1.854473
G7	0.88%	1.693909
G8	0.81%	1.608715
G9	0.80%	1.650192
G10	0.73%	1.622982
LS	-2.58%	-4.720989

Figure 1 Average monthly return

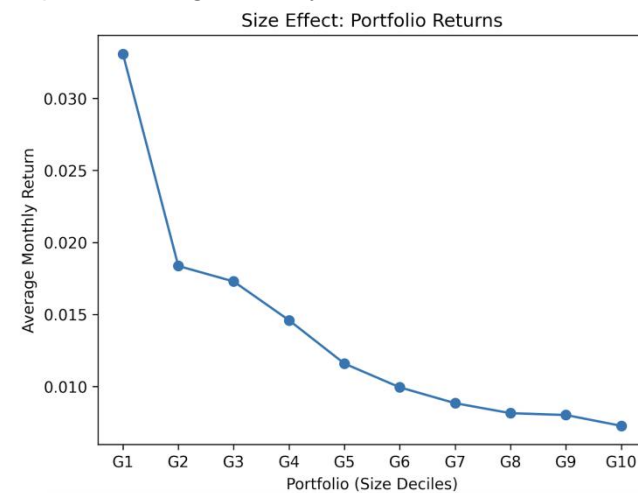


Table 2 CAPM regressions

Portfolio	Alpha	Alpha t	Beta_MKT	Beta t
G1	2.73%	5.012298	1.094739	14.369114
G2	1.26%	3.202619	1.09436	19.92991
G3	1.15%	3.321063	1.087976	22.407232
G4	0.88%	2.687769	1.091145	23.780804
G5	0.59%	1.893847	1.074968	24.696952
G6	0.42%	1.415053	1.089012	26.436468
G7	0.31%	1.147961	1.091037	29.302496
G8	0.24%	1.001048	1.083743	32.309956
G9	0.23%	1.161063	1.07679	38.69972
G10	0.17%	1.543362	1.054342	69.370441
LS	-2.56%	-4.663268	-0.040396	-0.525591

Size effect: Small-cap companies have higher average returns than large-cap companies

The Mean Return of LS is negative at -2.58%, showing the return of G1 is larger than G10. From G1 to G10, the mean return is decreasing while the company size is increasing, which follow the law of size effect.

The CAPM $\alpha = -2.56\%$, with $t = -4.66$, is significantly non-zero, suggesting that the size effect is economically and statistically significant.

So the results show **the existence of** size effect and market risk alone cannot fully explain the size effect.

b) Momentum effect

Table 3 The time-series average monthly return and t-value for each of the 5 portfolios

Portfolio	Mean Return	t-value
P1	1.70%	3.001718
P2	1.56%	2.978767
P3	1.45%	2.795891
P4	1.18%	2.319708
P5	0.62%	1.134988
LS	-1.07%	-3.61321

Figure 2 Average monthly return

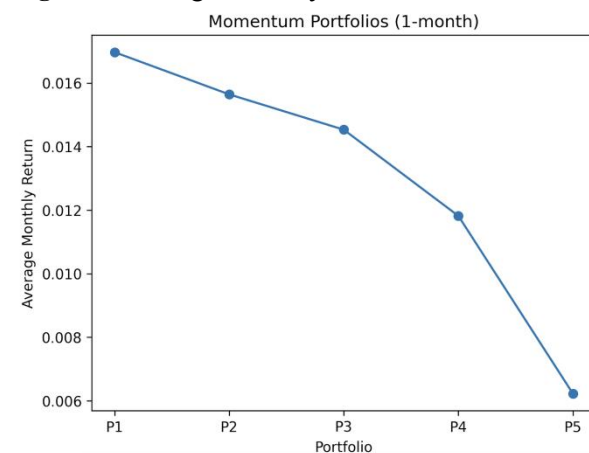


Table 4 Fama-French 3-factor regressions

Portfolio	Alpha	Alpha t	Beta_MKT	MKT_t	Beta_SMB	SMB_t	Beta_HML	HML_t
P1	0.72%	4.379101	1.033991	45.08834	0.873169	23.663099	-0.300828	-5.299808
P2	0.58%	4.890254	0.996319	60.335374	0.859317	32.341002	-0.117971	-2.886304
P3	0.47%	4.479236	1.006755	68.747396	0.834248	35.404111	-0.087278	-2.407871
P4	0.26%	2.253821	1.000833	63.056631	0.753895	29.519322	-0.133617	-3.401152
P5	-0.27%	-1.207368	1.00693	31.720925	0.728624	14.265139	-0.218248	-2.77774
LS	-1.00%	-3.330131	-0.027061	-0.649496	-0.144545	-2.156042	0.08258	0.800751

Momentum effect: Stocks that have performed well in the past often continue to perform well in the near future; conversely, stocks that have performed poorly often continue to perform poorly.

The results show that, the mean return at t is decreasing from P1 to P5, indicating the stocks that performed poorly will have better perform in the future. For LS, the mean return is -1.07%, with a of 1% and t-value of -3.33, showing the α is significantly non-zero.

For Fama regression, the MKT_Beta of LS is not significantly non-zero; the SMB_Beta of LS equals to -0.14, not far away from zero; the HML_Beta is not significantly non-zero. All these data show these 3 factors can't explain the LS alpha, means this situation is abnormal.

These results suggest there is **NOT** momentum effect.